
SR75 millimeter wave radar Communication Protocol

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Version information:

Date	Version	Version description
2022-04-12	1.0	Draft of SR75 communication protocol
2022-05-27	2.0	Add CANFD communication protocol description
2022-06-15	2.0	Modify Point Cloud Parse Sample Error
2022-08-11	3.0	Modifying Point Cloud Protocol, Parsing
2022-10-27	4.0	Add trace protocol, parsing,
2022-10-28	5.0	Add Trajectory Target Filter
2023-03-13	6.0	Add trace information output, configuration information input,
2023-04-23	7.0	Add the wrong frequency ID configuration, modify the target filtering mode table in Table 1-8, and modify the example parsing in 2.4 track status message
2024-01-24	8.0	Change the radar configuration and radar return memory distribution map, Modify point cloud data information protocol and parse

Note: Refer to "SR75 Radar Communication Protocol Iteration Information.xlsx" for detailed modification.

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1 Common configuration protocol

One of the radar sensor hardware interfaces is the UART-TTL (3.3 V level) interface. Generally, the default baud rate is 115200 (different types of radars can use their own default baud rates), with 8 data bits, 1stop bit, no parity bit, and no flow control. One is CAN, the default baud rate is 500K, standard CAN. Each data message starts with a start sequence and ends with a stop sequence. In each refresh cycle, the system status and target output status messages will be output. If the target is detected, the number of **targets detected in the target output status message is N**, where N is greater than 0. The target output status message will be followed by the target information message. The radar ID can be configured, and the output Message IDs can be changed.

The radar ID is $0 \sim 7$, and the frame ID calculation formula is as follows:

$$\text{MsgID} = \text{MsgID_0} + \text{SensorID} * 0x10$$

For example, if radar ID = 0, the frame ID is configured as 0 X 200; If radar ID = 1, the configuration frame ID is 0x210, and so on. After changing the radar ID, the radar will only respond to the new configuration message.

1.1 Universal serial port protocol frame format

1) Packet transmission format

Table 1- 1Packet transmission format

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x01	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

- ♦ Start code: 2-byte frame header-0xAA 0xAA.
- ♦ Frame ID: 2 bytes-the low byte is in the front, which shall be consistent with the radar ID.
- ♦ Command number: When the command is a write command, the most significant bit is 1. When the command is a read command, the most significant bit is 0.

- ◆ Parameters: unsigned numbers unless otherwise specified.
- ◆ End code: 2 bytes.

2) Radar reply format

Table 1-2Radar reply format

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0x81	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

- ◆ Start code: 2-byte frame header-0xAA 0xAA.
- ◆ Frame ID: 2 bytes-the low byte is in the front, which shall be consistent with the radar ID.
- ◆ Command number: when the command succeeds, the highest bit is 1; when the command fails, the highest bit is 0.
- ◆ Parameters: unsigned numbers unless otherwise specified.
- ◆ End code: 2 bytes.

1.2 General CAN protocol frame format

1) Packet transmission format

Table 1-3Packet transmission format

Frame ID	Command	Parameter						
0x0500	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
	0x81	0x00	0x00	0x00	0x00	0x00	0x00	0x00

- ◆ Frame ID: 2 bytes, consistent with radar ID.
- ◆ Command number: When the command is a write command, the most significant bit is 1. When the command is a read command, the most significant bit is 0.
- ◆ Parameters: unsigned numbers unless otherwise specified.

2) Radar reply format

Table 1-4Radar reply format

Frame ID	Command	Parameter						
0x0501	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8

	0x01	0x00	0x00	0x00	0x00	0x00	0x00	0x00
--	------	------	------	------	------	------	------	------

- ♦ Frame ID: 2 bytes, consistent with radar ID.
- ♦ Command number: when the command succeeds, the highest bit is 1; when the command fails, the highest bit is 0.
- ♦ Parameters: unsigned numbers unless otherwise specified.

Note that the CAN protocol frame does not have a header and trailer.

1.3 Common protocol frame ID

The frame ID (Message ID) is used to indicate the purpose of the message. The transmission direction of the frame ID includes radar receiving (In) and radar sending (Out). Each frame ID has a specific transmission direction.

Table 1- 5Common protocol frame ID

Num	In/Out	Message ID	Message Name	Comment
1	Receive	0x200	Sensor Configuration	Radar configuration (compatible with mainland)
2	Send	0x201	Sensor Back	Radar Return (Continental Compatible)
3	Receive	0x500	Sensor Configuration	Custom radar configuration
4	Send	0x501	Sensor Read Back	Custom Read Configuration Return
5	Send	0x502	Sensor Write Back	Custom write configuration return

1.3.1 Radar configuration (0x200)

This configuration command is compatible with the mainland configuration command (8 bytes of data are the same). This configuration command is a combined configuration command, and multiple parameters can be configured at the same time by one command.



Figure 1-1Radar Configuration (0x200) Memory Distribution Map

Table 1- 6Description of radar configuration (0x200)

Signal	Start	Len	Min	Max	Res	Unit
RadarCfg_MaxDistance_Valid	0	1	0	1	1	0x0:Invalid 0x1:Valid
RadarCfg_SensorID_Valid	1	1	0	1	1	0x0:Invalid 0x1:Valid
RadarCfg_RadarPower_Valid	2	1	0	1	1	0x0:Invalid 0x1:Valid
RadarCfg_OutputType_Valid	3	1	0	1	1	0x0:Invalid 0x1:Valid
RadarCfg_SendQuality_Valid	4	1	0	1	1	0x0:Invalid 0x1:Valid
RadarCfg_SendExtInfo_Valid	5	1	0	1	1	0x0:Invalid 0x1:Valid
RadarCfg_SortIndex_Valid	6	1	0	1	1	0x0:Invalid 0x1:Valid
RadarCfg_StoreInNvm_Valid	7	1	0	1	1	0x0:Invalid 0x1:Valid
RadarCfg_MaxDistance	22	10	0	2048	2	m
RadarCfg_StaggerFreqID	24	2	0	3	1	Error ID (0 ~ 3) (default ID is 0)
RadarCfg_SensorID	32	3	0	7	1	ID (0 ~ 7) (default ID is 0)
RadarCfg_OutputType	35	2	0	2	1	0x0:None 0x1:Objects 0x2: Clusters (default)
RadarCfg_RadarPower	37	3	0	7	1	0 x0: Standard (default)

CANFD_FrameType	40	1	0	1	1	0x0:CLASSIC Frame 0x1:FD Frame Point cloud version defaults to 0x1 Trace version defaults to 0x0 Modification is not supported at present
CANFD_FrameType_Valid	41	1	0	1	1	0x0:Invalid 0x1:Valid
RadarCfg_SendQuality	42	1	0	1	1	0x0:Inactive 0x1:Active
RadarCfg_SendExtInfo	43	1	0	1	1	0x0:Inactive 0x1:Active
RadarCfg_SortIndex	44	3	0	7	1	0 x0: No Sorting (default) 0 x1: Sort By Range (reserved)
RadarCfg_StoreNVM	47	1	0	1	1	0x0:Inactive 0x1:Active
RadarCfg_RCS_Threshold_Valid	48	1	0	1	1	0x0:Inactive 0x1:Active
RadarCfg_RCS_Threshold	49	3	0	7	1	0 x0: Standard (default) 0x1:High Sensitivity
RadarCfg_InterfaceType	52	2	0	3	1	0x0:UART 0x1:CAN Point cloud output does not support UART
RadarCfg_InterfaceSelectValid	54	1	0	1	1	0x0:Inactive 0x1:Active
RadarCfg_LVDS Valid	55	1	0	1	1	0x0:Inactive 0x1:Active
RadarCfg_LVDS Select	56	1	0	1	1	0x0:Inactive 0x1:Active Temporarily closed
CANFD_DataBaudRate	57	3	0	7	1	CANFD Data Field Baud Rate (Point Cloud), 0x0:1M 0 x1: 4M (default) 0x2:2M Modification is not supported

						at present
RadarCfg_BaudRate_Valid	60	1	0	1	1	0x0:Inactive 0x1:Active
RadarCfg_BaudRate	61	3	0	7	1	When it is CAN, 0x0:500K 0x1:250K 0x2: 1M (default) When it is a serial port, 0 x0: 115 200 (default) 0x1:9600 0x2:921600

For example: AA 00 02 82 00 01 80 00 55 (change the radar ID to 1 and save it).

1.3.2 Configuration return (0x201)

When the radar receives the 0x200 configuration command, it needs to return this command within 1 second. This instruction is compatible with the mainland configuration return instruction (8 bytes of data are the same).

Example: AA AA 01 02 00 55 55

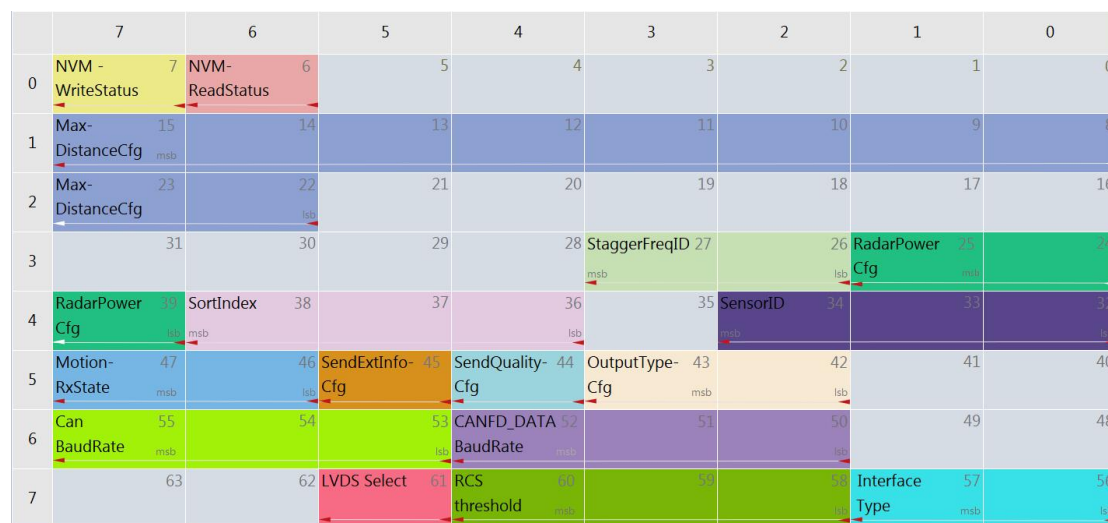


Figure 1-2Radar configuration returns (0x201) memory distribution map

Table 1- 7Description of radar configuration return (0x201)

Signal	Start	Len	Min	Max	Res	Uint
RadarState_NVMReadStatus	6	1	0	1	1	0x0:Failed 0x1:Successful

RadarState_NVMWriteStatus	7	1	0	1	1	0x0:Failed 0x1:Successful
RadarState_MaxDistanceCfg	22	10	0	2046	2	m
RadarState_StaggerFreqID	26	2	0	3	1	Current wrong frequency ID (0 ~ 3)
RadarState_SensorID	32	3	0	7	1	Current radar ID (0 ~ 7)
RadarState_SortIndex	36	3	0	7	1	0x0:nosorting 0 x1: (reserved) 0x2: (reserved)
RadarState_RadarPowerCfg	39	3	0	7	1	0x0:Standard
RadarState_OutputTypeCfg	42	2	0	3	1	0x0:None 0x1:Object 0x2:Cluster
RadarState_SendQualityCfg	44	1	0	1	1	0x0:Inactive 0x1:Active
RadarState_SendExtInfoCfg	45	1	0	1	1	0x0:Inactive 0x1:Active
RadarState_MotionRxState	46	2	0	3	1	0x0:input ok 0x1:speed missing 0x2:yaw rate missing 0x3:speed and yaw rate missing
CANFD_FrameType	49	1	0	1	1	0x0:CLASSIC Frame 0x1:FD Frame
CANFD_DataBaudRate	50	3	0	7	1	0x1: 4M (default) when CANFD
RadarState_BaudRate	53	3	0	7	1	0x2: 1M in case of CAN (default) In case of serial port, 0x0: 115200 (default)
RadarCfg_InterfaceType	56	2	0	3	1	0x0:UART 0x1:CAN Point cloud output does not support UART
RadarState_RCS_threshold	58	3	0	7	1	0x0:Standard 0x1:high sensitivity

1.3.3 Target Filtering Configuration (0x202)

Filter mode and filter condition can be configured through (0x202). Target filtering methods include speed filtering, angle filtering and distance filtering, which are mutually exclusive. And filter that track target, outputting the track target if the track target meet the filtering conditions, and filtering the track target if the track target do not meet the filtering conditions.

The target filtering mode refers to the 408 protocol, and the FilterCfg _ Index parameter is used to select the filtering condition. C and O respectively specify whether the target filtering mode can be used for Cluster or Object (X indicates support).

Table 1- 8Target filter mode table

Index	Filtering mode	C	O	Description
0x1	Distance		x	Radial distance in m, range [0,90]
0x2	Azimuth		x	Angle, supports 2 angle ranges in °, range: [-90, 90]
0x3	Velocity		x	Radial velocity in km/H in the range [-60, 60]
0x1F	FilterClear	x	x	When Filter Cfg _ Active = 0 and Filter Cfg _ Valid = 1, all current filtering modes are cleared

In the memory distribution diagram of target filtering configuration (0x202), FilterCfg _ Min _ Para and FilterCfg _ Max _ Para respectively represent the minimum and maximum values of filtering parameters of different filtering modes. Refer to the description table of target filtering configuration (0x202).

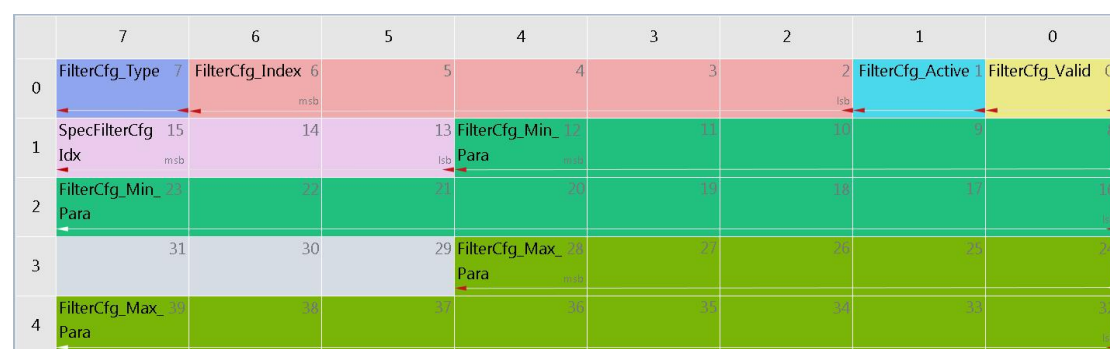


Figure 1- 3Target Filtering Configuration (0x202) Memory Distribution Map

Table 1- 9Target filtering configuration (0x202) description

Signal	Start	Len	Offset	Min	Max	Res	Uint
FilterCfg_Valid	0	1		0	1	1	0x0:Invalid 0x1:Valid
FilterCfg_Active	1	1		0	1	1	0x0:Inactive 0x1:Active
FilterCfg_Index	2	5		0	31	1	For the filtering mode, see the target filtering mode table.
FilterCfg_Type	7	1		0	1	1	0x0:Cluster filter 0x1:Object filter

SpecFilterCfg_Idx	13	3		0	1	1	Filter ID for a specific filter mode
FilterCfg_Min_Distance	16	13		0	1638. 2	0.2	m
FilterCfg_Max_Distance	16	13		0	1638. 2	0.2	m
FilterCfg_Min_Azimuth	16	13	-90	-90	1629 2	2	Degrees, two angular ranges can intersect
FilterCfg_Max_Azimuth	16	13	-90	-90	1629 2	2	Degree
FilterCfg_Min_Velocity	16	13	-60	-60	8131	1	Km/H, speed must be set continuously, range [-60 km/H, 60 km/H]
FilterCfg_Max_Velocity	16	13	-60	-60	8131	1	Km/H, speed must be set continuously, range [-60 km/H, 60 km/H]

- 1) Filter Cfg _ Valid: If true, the filtering configuration can be updated.
- 2) Filter Cfg _ Active: enable or disable the corresponding filtering mode.
- 3) Filter Cfg _ Index: Specify the filtering method to be configured.
- 4) SpecFilter Cfg _ Idx: The different configuration ids for a particular filter. For example, if Angular filtering supports two ranges, you can use the Spec FilterCfg _ Idx to specify the range ID as 0 or 1.
- 5) FilterCfg _ Max _ Para, FilterCfg _ Min _ Para: Configure minimum and maximum values for filtering.

1.3.4 Target filtering return (0x203)

After configuring the filtering parameters, the radar replies the filtering status messages of FilterState _ Header (0x203) and FilterState _ Cfg (0x204) with modified filtering mode.

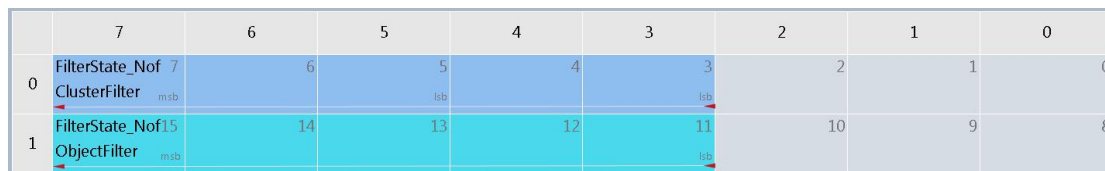


Figure 1-4 Target Filtering Returns (0x203) Memory Distribution Map

Table 1-10 Target filtering return (0x203)

Signal	Start	Len	Offset	Min	Max	Res	Uint
FilterState_NofClusterFilter	3	5		0	31	1	Point Cloud Schema Configured Filter Quantity

FilterState_NoObjectFilter	11	5		0	31	1	Trace Mode Configured Filter Quantity
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1.3.5 Target filtering return (0x204)

After the filter parameters are configured, the radar replies with the filtering status message of FilterState _ Header (0x203) and FilterState _ Cfg (0x204) with the filtering mode modified. For the filtering status message, refer to the description of target filtering configuration (0x202).

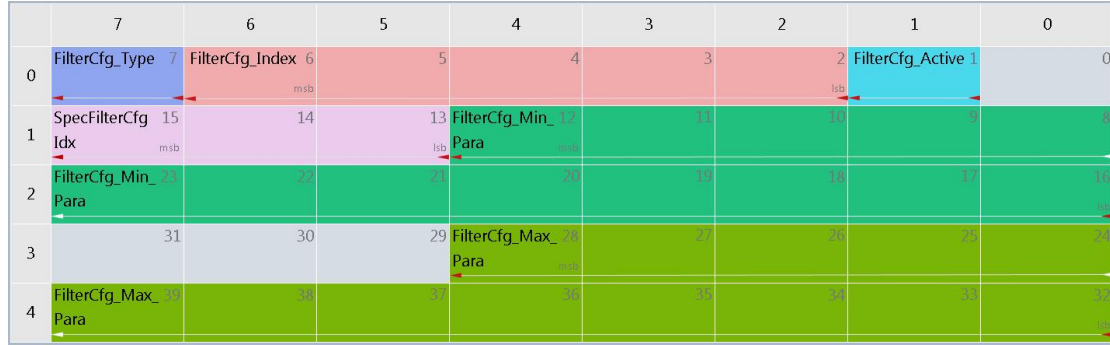


Figure 1-5 Target Filters Return (0x204) Memory Distribution Map

1.3.6 Collision Detection Configuration (0x400)

The CollDetCfg (0x400) message is used to activate area-based collision detection. After the collision detection function is activated, the radar sends CollDetState (0x408) and the message CollDetRegionState (0x402) periodically (10s) for all configured areas. Currently, only one zone is supported to be set, and the zone ID is 1. The radar is in objects mode by default, and the collision detection function is not enabled.

Only Objects mode has collision detection, that is, the RadarCfg _ OutputType is set to Objects (0x1). When collision detection is enabled.



Figure 1-6 Collision Detection Configuration (0x400) Memory Distribution Map

Table 1- 11 Collision detection area setting (0x401) description

Signal	Start	Len	Min	Max	Res	Uint
ColDetCfg_WarningReset	0	1	0	1	1	0x0:idle 0x1:reset warnings
ColDetCfg_Activation	1	1	0	1	1	0x0:Inactive 0x1:Active
ColDetCfg_MinTime_Valid	3	1	0	1	1	0x0:Inactive 0x1:Active

ColDetCfg_ClearRegions	7	1	0	1	1	0x0:idle 0x1:clear regions
ColDetCfg_MinTime	8	8	0.0	25.5	0.1	Seconds (sec)

Table 1- 12 Collision detection area setting (0x401) description

Signal	Start	Description
ColDetCfg_WarningReset	0	0 x1: Resets the current warning for all zones when enabled 0x0: Turn on the warning mode of all current zones when disabled
ColDetCfg_Activation	1	Activate/deactivate collision detection
ColDetCfg_MinTime_Valid	3	When true, the time parameter is allowed to be changed
ColDetCfg_ClearRegions	7	When true, clears all zone configuration
ColDetCfg_MinTime	8	The minimum time an object in the area needs to be detected before a warning is triggered

- 1) ColDetCfg _ ClearRegions: Clears the current area. Send 0x400 0 X80 0 x00, the current area is cleared, and the radar no longer outputs 0 x408 0 x402 0 x60E.
- 2) To open a zone, set the ColDetCfg _ Activation to 1.
- 3) ColDetCfg _ Warning Reset: 0x1: reset the current activation warning of all areas when enabled; 0x0: Turn on the warning mode of all current defense zones when disabled
- 4) ColDetCfg _ MinTime _ Valid: Not used yet.
- 5) ColDetCfg _ MinTime: Not used yet.

1.3.7 Collision Detection Area Setting (0x401)

The Collision detection region configuration (0x401) supports the setting of 3 detection areas, and each area is formed by expanding the abscissa and ordinate of two points, as shown in the following figure:

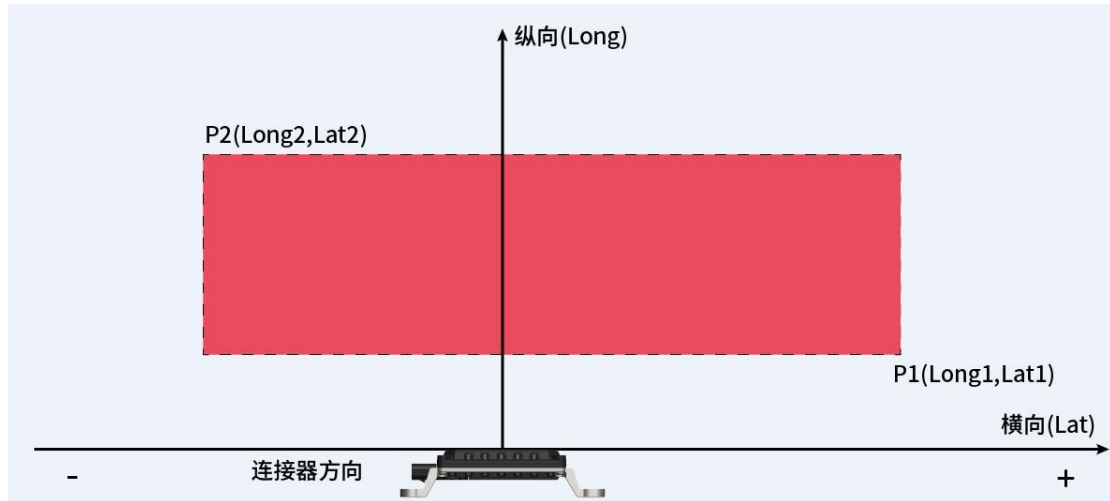


Figure 1-7 Schematic Diagram of Collision Detection Area Coordinate Settings

When configuring the collision detection area (0x401), make sure that the collision detection configuration (0x400) has been configured, and make sure that the ColDetCfg _ Activation = 1 and the ColDetCfg _ ClearRegions = 0; otherwise, the area cannot be configured.

	7	6	5	4	3	2	1	0
0	ColDetRegCfg 7 CoordinatesValid	ColDetRegCfg 6 Activation	5	4	ColDetRegCfg 3 Warning	ColDetRegCfg 2 RegionID	1	0
1	Max_Output Number	15	14	13	12	11	10	9
2	ColDetRegCfg 23 Point1Long	22	21	20	19	18	17	16
3	ColDetRegCfg 31 Point1Long	30	29	28	27	ColDetRegCfg 26 Point1Lat	25	24
4	ColDetRegCfg 39 Point1Lat	38	37	36	35	34	33	32
5	ColDetRegCfg 47 Point2Long	46	45	44	43	42	41	40
6	ColDetRegCfg 55 Point2Long	54	53	52	51	ColDetRegCfg 50 Point2Lat	49	48
7	ColDetRegCfg 63 Point2Lat	62	61	60	59	58	57	56

Figure 1-8 Collision Detection Locale (0x401) Memory Distribution Map

Table 1- 13 Collision Detection Area Status (0x402) Description

Signal	Start	Len	Offset	Min	Max	Res	Uint
ColDetRegCfg_RegionID	0	3		0	7	1	ID number can only be 1-3
ColDetRegCfg_Warning	3	1		0	1	1	0x0:Inactive 0x1:Active
ColDetRegCfg_Activation	6	1		0	1	1	0x0:Inactive 0x1:Active
ColDetRegCfg_CoordinatesValid	7	1		0	1	1	0x0:Invalid 0x1:Valid

ColDetRegCfg _MaxOutputNumber	8	8		0	255	1	Supports up to 128 targets
ColDetRegCfg _Point1Long	27	13	-500	-500	1138.2	0.2	Unit m
ColDetRegCfg _Point1Lat	32	11	-204.6	-204.6	204.8	0.2	Unit m
ColDetRegCfg _Point2Long	51	13	-500	-500	1138.2	0.2	Unit m
ColDetRegCfg _Point2Lat	56	11	-204.6	-204.6	204.8	0.2	Unit m

- 1) ColDetRegCfg _RegionID: support drawing three areas, and the defense area ID is 1 ~ 3.
- 2) ColDetRegCfg _ Warning: zone alarm mode, when enabled, the target in the defense zone outputs 0x60E alarm message.
- 3) ColDetRegCfg _ Activation: whether the collision detection function is activated, 0x0: inactive, 0x1: active. After the function is activated and the coordinates are valid, the subsequent rectangular frame settings will take effect, that is, the last set parameters are valid when the power is turned on again.
- 4) ColDetRegCfg _ CoordinatesValid: The coordinate point setting 0x0: invalid is invalid, and 0x1: active is valid. Only when the collision detection function is activated and the enabled coordinate is valid, the coordinate setting will take effect, otherwise it will not take effect.
- 5) ColDetRegCfg _ Point1 Long: longitudinal distance value of coordinate point 1;
- 6) ColDetRegCfg _ Point1Lat: horizontal distance value of coordinate point 1;
- 7) ColDetRegCfg _ Point2 Long: Longitudinal distance value of coordinate point 2;
- 8) ColDetRegCfg _ Point1Lat: horizontal distance value of coordinate point 2;
- 9) Note for setting coordinate point: coordinate point 1 is the coordinate value of the lower right corner of the rectangular box, and coordinate point 2 is the coordinate value of the upper left corner of the rectangular box. I.e. satisfied,
ColDetRegCfg _Point1Long < ColDetRegCfg _Point2Long
ColDetRegCfg _Point1Lat > ColDetRegCfg _Point1Lat
Otherwise, the set rectangle is invalid and is not saved in Flash.

1.3.8 Collision Detection Area Status (0x402, 0x405/0x406)

When either of the ColDetRegCfg _ Activation and ColDetRegCfg _ CoordinatesValid in 0x401 is not 1, the radar will only output the objects mode target, and the defense area will not take effect.

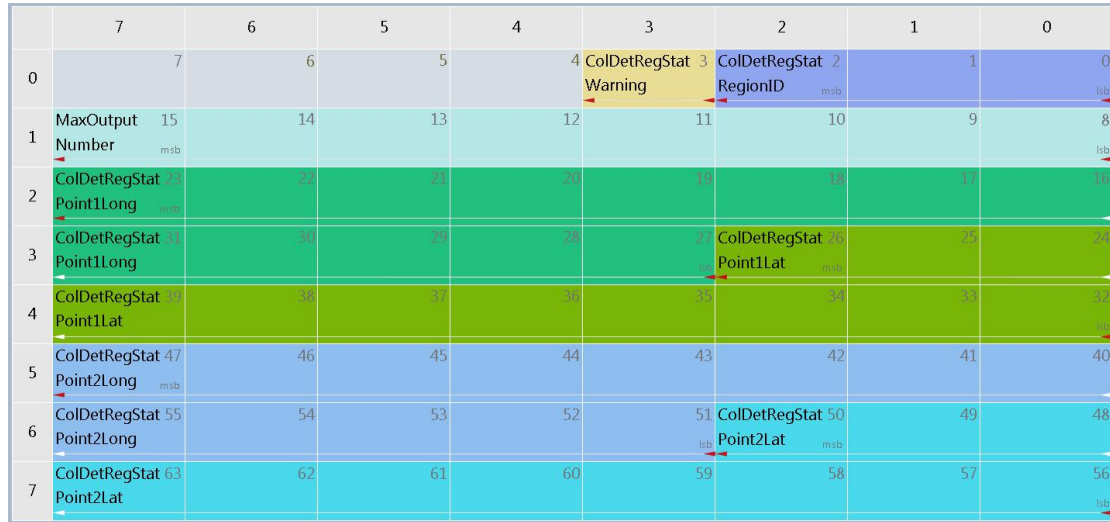


Figure 1- 9Collision Detection Area Status (0x402) Memory Distribution Map

Table 1- 14Collision Detection Area Status (0x402) Description

Signal	Start	Len	Offset	Min	Max		Res	Unit
ColDetRegStat_RegionID	0	3		0	7		1	Region number of the current region
ColDetRegStat_Warning	3	1		0	1		1	
ColDetRegStat_MaxOutputNumber	8	8		0	255		1	
ColDetRegStat_Point1Long	27	13	-500	-500	1138.2		0.2	Unit m
ColDetRegStat_Point1Lat	32	11	-204.6	-204.6	204.8		0.2	Unit m
ColDetRegStat_Point2Long	51	13	-500	-500	1138.2		0.2	Unit m
ColDetRegCfg_Point2Lat	56	11	-204.6	-204.6	204.8		0.2	Unit m

- 1) ColDetRegStat_Warning: warning of whether the area is open.
- 2) ColDetRegStat_RegionID: the region number of the current region, which can only be 1 ~ 3.
- 3) ColDetRegStat_MaxOutputNumber: The maximum number of targets supported in the current area that meet the collision detection conditions. If the number exceeds the maximum number, ColDetRegStat_MaxOutputNumber targets with a short radial distance will be taken.

Send 0x405 command to read the collision detection area status, 0x405 0xXX 0x00 0x40x40.

1.3.9 Collision Detection Status (0x408)

Activate area-based collision detection via ColDetCfg (0x408) message. When the collision detection function is activated, the sensor periodically sends the current collision detection configuration and warning status via message ColDetState (0x408). For a single area, the radar sends ColDetRegState (0x402).

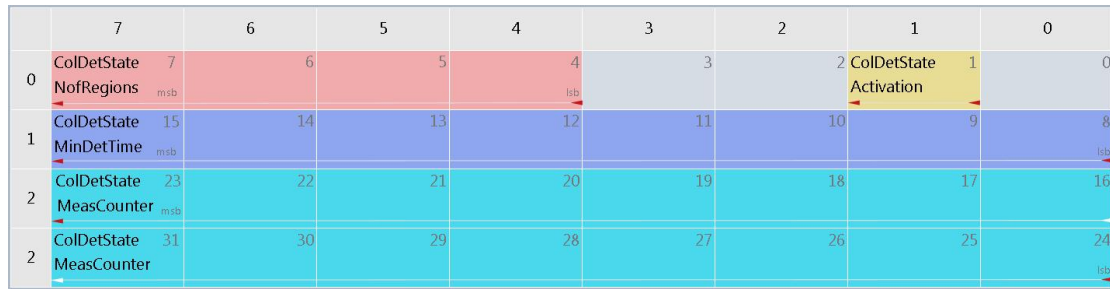


Figure 1- 10 Collision Detection Status (0x408) Memory Distribution Map

Table 1- 15 Collision Detection Status (0x408) Description

Signal	Start	Len	Min	Max	Res	Unit
ColDetState_Activation	1	1	0	1	1	0x0:Inactive 0x1:Active
ColDetState_NofRegions	4	4	0	15	1	
ColDetState_MinDetTime	8	8	0.0	25.5	0.1	Seconds/sec
ColDetState_MeasCounter	24	16	0	65535	1	

Table 1- 16 Collision Detection Status (0x408) Description

Signal	Start	Description
ColDetState_Activation	1	Collision detection active state
ColDetState_NofRegions	4	Number of configured zones
ColDetState_MinDetTime	8	The minimum detection time of the target before the currently configured warning departure. Not used yet
ColDetState_MeasCounter	24	Measurement cycle count (cyclic accumulation, reset to 0 when greater than 65535, continue accumulation)

1.3.10 Collision detection area alarm (0x60E)

When the collision detection configuration warning is disabled (ColDetCfg _ Warning Reset = 0) and the collision detection area warning mode is enabled (ColDetRegCfg _ Warning = 1), the radar sends a message Obj _ Warning (0x60E) to each target in the defense area. This message indicates that the current target is violating an area.

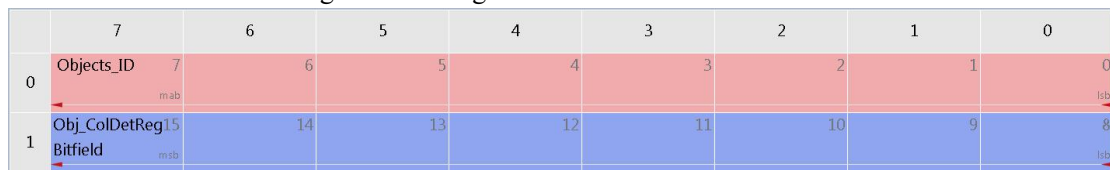


Figure 1- 11 Collision Detection Area Alarm (0x60E) Memory Distribution Map

Table 1- 17Description of collision detection zone alarm (0x60E)

Signal	Start	Len	Min	Max	Res	Uint
Objects_ID	0	8	0	255	1	Target ID within the area
Obj_ColDetRegBitfield	16	8	0	255	1	Bit field of the region in which the target is set to the region ID of the region

1.3.11 Custom configuration

1.3.11.1 Radar model (read-only)

Table 1- 18Read radar model command

Start Code		Frame ID Low position	Frame ID high order bits	Read Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xA	0xA	0x00	0x05	0x01	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 19Command format returned after reading radar model

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0x81	0xFF	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Reply according to the following table, which can be supplemented.

Table 1- 20Radar Model Enumeration Type Table

Data [1] enumeration values	Radar type
0x00	SR60
0x01	ETR10
0x02	MR76
0x03	MR76S
0x04	MR62
0x05	TGR20

0x06	TCR300
0x07	SR73F
0x09	SR75

1.3.11.2 Radar Firmware Version (Read Only)

Table 1-21Read radar version command

Start Code		Frame ID Low	Frame ID high order bits	Read Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x02	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1-22Command format returned after reading radar version

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0x82	0xFF	0xFF	0xFF	0x00	0x00	0x00	0x00	0x55	0x55

Reply according to the following table, which can be supplemented.

Table 1-23Radar firmware version example table

Data[1]~Data[3]	Radar firmware version
0x01 0x00 0x00	V1.0.0
0x02 0x01 0x03	V2.1.3

1.3.11.3 Radar ID, wrong frequency ID

Table 1-24Read radar ID command

Start Code		Frame ID Low	Frame ID high order bits	Read Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x04	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 25Command format returned by reading radar ID

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0x84	0xFF	0xFF	0x00	0x00	0x00	0x00	0x00	0x55	0x55
New start code		Frame ID Low	Frame ID high order bits	Write Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x84	0xFF	0xFF	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 26Command format returned by writing radar ID

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x02	0x05	0x84	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Reply according to the following table. It can be added that the new ID can take effect only after it is set and restarted.

Table 1- 27Radar ID, Error Frequency ID Definition Table

Data[1]	Radar ID	Data[2]	Wrong frequency ID
0x00	0	0x00	0
0x01	1	0x01	1
...	...	0x02	2
0x07	7	0x03	3

1.3.11.4 Radar output interface

Table 1- 28Read radar output interface command

Start Code		Frame ID Low	Frame ID high order bits	Read Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x05	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 29Read command format returned from radar output interface

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0x85	0xFF	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Table 1- 30Command for writing radar output interface

Start Code		Frame ID Low	Frame ID high order bits	Write Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x85	0xFF	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 31Command format returned by writing radar output interface

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x02	0x05	0x85	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Reply according to the following table, which can be supplemented.

- 1) It is modified that the radar output interface needs to be restarted to take effect, and the modified baud rate is 1.3.3.6.Default baud rate;

2) Uart and CAN must be selected, and only one of them can be selected;

Table 1- 32Radar output interface definition table

Data[1]	Radar output interface
0x00	Uart
0x01	CAN

1.3.11.5 Radar hardware version (read-only)

Table 1- 33Read radar hardware version command

Start Code		Frame ID Low	Frame ID high order bits	Read Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x06	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 34Command format returned after reading radar hardware version

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0x86	0xFF	0xFF	0xFF	0x00	0x00	0x00	0x00	0x55	0x55

Reply according to the following table, which can be supplemented.

Table 1- 35Radar hardware version example table

Data[1]~Data[3]	Radar hardware version
0x01 0x00 0x00	V1.0.0
0x03 0x00 0x00	V3.0.0

1.3.11.6 Radar baud rate

Table 1- 36Read radar baud rate command

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13

	1	2	3	4	5	6	7	8	9	10	11	12	13
0xAA	0xAA	0x00	0x05	0x09	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 37Command format returned by reading radar baud rate

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0x89	0xFF	0xFF	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Table 1- 38Write radar baud rate command

Start Code		Frame ID Low	Frame ID high order bits	Write Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x89	0xFF	0xFF	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 39Command format returned by writing radar baud rate

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x02	0x05	0x89	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Reply according to the following table, which can be supplemented.

And 1.3.3.4[Radar interface](#)Used together, the switch needs to be restarted to take effect.

Table 1- 40 Uart baud rate definition table

Data[1]	Uart baud rate
0x00	115 200 (default)
0x01	9600
0x02	921600
0x03	57600

Table 1- 41 CAN Baud Rate Definition Table

Data[1]	Can baud rate
0x00	500K
0x01	250K
0x02	1M (default)
0x03	125K

Table 1- 42 CANFD Baud Rate Definition Table

Data[1]	CANFD arbitration domain baud rate
0x00	500K
0x01	250K
0x02	1M (default)
0x03	125K
Data[2]	CANFD Data Field Baud Rate
0x00	1M
0x01	5M (default)
0x02	2M
0x03	500K

1.3.11.7 Radar activation status (read-only)

Table 1- 43Read radar active status command

Start Code		Frame ID Low	Frame ID high order bits	Read Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x0F	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 44Command format returned after reading radar activation status

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0x8F	0xXX	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Reply according to the following table, which can be supplemented.

Table 1-45Activation state definition table

Data[1]	Uart baud rate
0x00	Activated
0x01	Not activated

1.3.11.8 Radar installation parameters

Table 1-46Read radar installation parameter command

Start Code		Frame ID Low	Frame ID high order bits	Read Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x12	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1-47Read the command format returned by radar installation parameter

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0x92	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0x00	0x55	0x55

Table 1-48Command for writing installation parameters of radar

Start Code		Frame ID Low	Frame ID high order bits	Write Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x92	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1-49Command format returned by writing radar installation parameter

Start Code		Frame ID Low	Frame ID high order	Life Ling	Parameter							End Code	
------------	--	--------------	---------------------	-----------	-----------	--	--	--	--	--	--	----------	--

		bits											
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x02	0x05	0x92	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Reply according to the following table, which can be supplemented.

Calculation formula: actual height (unit: m) * 100 = installation height

Actual angle in degrees * 100 = mounting angle

The installation horizontal angle and the installation pitch angle are of the signed type.

Table 1- 50 Installation parameter definition table

Data[1]-Data[7]	Status
0x00	Install height high byte and height compensation
0x01	Installation Height Low Byte
0x02	Install Horizontal Angle High Byte
0x03	Install Horizontal Angle Low Byte
0x04	Install the high byte of the pitch angle
0x05	Install the low byte of the pitch angle
0x06	Angle compensation

Data [0] 0-2bit represents the high byte of installation height, and 3-7bit represents the height compensation value.

Where the 7th bit represents the sign, 0 is positive and 1 is negative. 3-6 bits means that the compensation of 0-15 meters is equal to the compensation of the actual height * 10.

For example, Data [1]-Data [2] is 0xA2 0x58, the compensation height is -0.4m, and the actual height is 6m.

Data [7] 0-3bit represents the installation of pitch angle compensation, the third bit represents the sign, 0 is positive, 1 is negative.

0-2 bits, each bit represents 1 ° of compensation.

For example, Data [7] is 0x0C for -4 ° compensation.

Data [7] 4-7bit represents horizontal angle compensation. The seventh bit represents the sign, with 0 being positive and 1 being negative.

Each of the 4th to 6th bits represents a compensation of 1 °.

For example, Data [7] is 0xC0 for -4 ° compensation.

1.3.11.9 Radar Area Target Filtering Mode

Table 1- 51 Read target out of radar area mode command

Start Code	Frame ID Low	Frame ID high	Read Life Ling	Parameter	End Code
------------	--------------	---------------	----------------	-----------	----------

			order bits										
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x13	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 52Command format returned by reading target mode out of radar area

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0x93	0xFF	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Table 1- 53Command for writing target mode outside radar area

Start Code		Frame ID Low	Frame ID high order bits	Write Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x93	0xFF	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 54Command format returned by writing target mode out of radar area

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x02	0x05	0x93	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Reply according to the following table. It can be supplemented. It needs to be restarted to take effect.

Table 1- 55Silent mode definition table

Data[1]	Status
0x00	Out-of-area target display

0x01	Out-of-area targets are not displayed
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1.3.11.10 Radar soft restart (write-only)

Table 1- 56Write radar soft restart command

Start Code		Frame ID Low	Frame ID high order bits	Write Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0xFF	0x10	0x01	0x02	0x03	0x04	0x05	0x06	0x55	0x55

Restart after writing, no reply.

1.3.11.11 Radar detection range

Table 1- 57Read radar detection range command

Start Code		Frame ID Low	Frame ID high order bits	Read Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x14	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 58Command format returned by reading radar detection distances

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0x94	0xFF	0xFF	0xFF	0xFF	0x00	0x00	0x00	0x55	0x55

Table 1- 59Write radar detection distance command

Start Code		Frame ID Low	Frame ID high order bits	Write Life Ling	Parameter							End Code	
Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte

0	1	2	3	4	5	6	7	8	9	10	11	12	13
0xAA	0xAA	0x00	0x05	0x94	0xFF	0xFF	0xFF	0xFF	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1-60 Command format returned by writing radar detection distance

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x02	0x05	0x94	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Reply according to the following table, which can be supplemented.

Calculation formula: actual maximum detection distance (unit: m) * 100 = maximum detection distance (50 ~ 600)

Actual minimum detection distance (unit: m) * 100 = minimum detection distance (0 ~ 50)

Actual longitudinal distance correction (unit: m) * 100 = longitudinal distance correction (-200 ~ 200)

Table 1-61 Radar detection distance definition table

Data[1]~Data[4]	Status
0x00	Maximum detection distance high bit
0x01	Maximum detection distance low bit
0x02	Minimum detection distance high order
0x03	Minimum detection distance low bit
0x04	Longitudinal distance correction (signed-type) high order
0x05	Longitudinal distance correction (signed-type) low order

1.3.11.12 Radar detection range

Table 1-62 Read radar detection range command

Start Code		Frame ID Low	Frame ID high order bits	Read Life Ling	Parameter							End Code	
Byte 0	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte

	1	2	3	4	5	6	7	8	9	10	11	12	13
0xAA	0xAA	0x00	0x05	0x15	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 63Command format returned by reading radar detection range

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0x95	0xFF	0xFF	0xFF	0xFF	0x00	0x00	0x00	0x55	0x55

Table 1- 64Command for writing radar detection range

Start Code		Frame ID Low	Frame ID high order bits	Write Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x95	0xFF	0xFF	0xFF	0xFF	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 65Command format returned by writing radar detection range

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x02	0x05	0x95	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Reply according to the following table, which can be supplemented.

Calculation formula: actual left detection range (unit: m) * 100 = left detection range (0 ~ 100)

Actual right detection range (unit: m) * 100 = right detection range (0 ~ 100)

Table 1- 66Radar Detection Range Definition Table

Data[1]~Data[4]	Status
0x00	Left detection range high order
0x01	Left detection range low order

0x02	Right detection range high order
0x03	Right detection range low order

1.3.11.13 Radar sensitivity and mode selection

Table 1- 67Read radar sensitivity and mode select command

Start Code		Frame ID Low	Frame ID high order bits	Read Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x16	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 68Read command format returned by radar sensitivity and mode selection

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0x96	0xXX	0xXX	0xXX	0xXX	0x00	0x00	0x00	0x55	0x55

Table 1- 69Write radar sensitivity and mode selection command

Start Code		Frame ID Low	Frame ID high order bits	Write Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x96	0xXX	0xXX	0xXX	0xXX	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 70Write command format returned by radar sensitivity and mode selection

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13

	1	2	3	4	5	6	7	8	9	10	11	12	13
0xAA	0xAA	0x02	0x05	0x96	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Reply according to the following table, which can be supplemented.

Calculation formula: actual rod dropping time (unit: second) * 100 = rod dropping time (50 ~ 800)

Sensitivity range: 0 ~ 7

Table 1-71 Radar sensitivity and mode selection definition table

Data[1]~Data[7]	Status
0x00	Control switch, multiple can be opened at the same time 0 x01 sensitivity switch 0x02 Operation Mode Switch 0x04 Man-vehicle partition switch 0 x08 Detect Direction Switch 0x10 Rod drop time switch
0x01	Sensitivity
0x02	Working mode The 0 x00 advertising pole is on the left side of the radar The 0x01 advertising pole is on the right side of the radar. 0 x02 Camera Obscura Mode 0 x10 single lever model 0x11 trigger mode
0x03	Distinguish between people and vehicles 0 x00 is 0 x01 No
0x04	Detect the direction 0 x00 is not direction sensitive 0 x01 left to right 0 x02 right to left
0x05	Rod drop time high
0x06	Rod drop time low

1.3.11.14 Radar count statistics

Table 1-72 Read radar count statistics command

Start Code	Frame ID	Frame ID	Read Life	Parameter	End Code
------------	----------	----------	-----------	-----------	----------

		Low	high order bits	Ling									
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x17	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 73Command format returned by reading radar counting statistics

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0x97	0xFF	0xFF	0xFF	0xFF	0x00	0x00	0x00	0x55	0x55

Table 1- 74Write clearing command for radar counting statistics

Start Code		Frame ID Low	Frame ID high order bits	Write Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x97	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 75Command format for writing radar counting statistics clearing return

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x02	0x05	0x97	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Reply according to the following table, which can be supplemented.

Table 1- 76Radar sensitivity and mode selection definition table

Data[1]~Data[4]	Radar count statistics
-----------------	------------------------

1.3.11.15 Radar filtering mode

Table 1- 77Read command format of current filtering mode of radar

Start Code		Frame ID Low	Frame ID high order bits	Read Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x26	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 78Read the command format returned by the radar filter mode

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0xA6	0xFF	0x00	0x00	0x00	0x00	0x00	0x00	0x55	0x55

Reply according to the following table, which can be supplemented.

Table 1- 79Radar filtering mode

Data[1]	Radar filtering mode
0x01	Distance
0x02	Azimuth
0x03	Velocity
0xFF	None

1.3.11.16 Radar filtering parameters

Table 1- 80Read radar filter parameter command

Start Code		Frame ID Low	Frame ID high order bits	Read Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x00	0x05	0x27	0xFF	0xFF	0x00	0x00	0x00	0x00	0x00	0x55	0x55

- 1) Byte 5 is the filter to be read.
- 2) Byte 6 is the SpecFilterCfg _ Idx of the filtering mode to be read. Refer to the target filtering configuration (0x202).

The format of the frame returned by the radar after executing the command is shown in the following table:

Table 1- 81Read the command format returned by the radar filter parameter

Start Code		Frame ID Low	Frame ID high order bits	Life Ling	Parameter							End Code	
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13
0xAA	0xAA	0x01	0x05	0xA7	0xFF	0xFF	0xFF	0xFF	0xFF	0x00	0x00	0x55	0x55

Filter parameter return refers to the target filter return (0x204), and byte 5 to byte 9 correspond to byte 0 to byte 4 of the target filter return (0x204).

2 Common Target Output Protocol (Continental Compatible)

This protocol applies to the 77 G radar, model SR75.

The frame format of the target output protocol in this subsection conforms to the frame format shown in the data packet sending format table, and the target output frame ID and data content are compatible with the Continental Objects target output protocol. The list of target output frame IDs is as follows:

Table 2- 1 Message ID definition

Num	In/Out	Message ID	Message Name	Comment
1	Send	0x201	Sensor Status	Radar status message
2	Send	0x700	Version Info	Radar version message
3	Send	0x60A	Target Status	Target status messages
4	Send	0x60B	Target Info	Target message

For the frame ID content definition in the above table, please refer to the Continental Agreement.

Output period:

The output period of 0x201 and 0x700 radar status messages is 10 seconds;

When CAN is used to output targets, CAN and CANFD protocols can be used. One frame of data of CAN protocol is 8 bytes, that is, the data information of one target, and one frame of data of CANFD protocol can be up to 64 bytes, that is, the data information of 8 targets. The CAN box can only output CAN protocol data, and the CANFD box can output both CAN protocol data and CANFD protocol data. Examples are as follows.

序号	时间标识	源通道	帧ID	帧类型	帧格式	CAN类型	方向	长度	数据
13	2022-04-12 08:50:03.913	Ch1	0x600	标准帧	数据帧	CAN	Rx	8	00 01 00 0A 00 01 00 00
14	2022-04-12 08:50:03.913	Ch1	0x701	标准帧	数据帧	CAN	Rx	8	04 4F 29 F7 80 20 10 04
15	2022-04-12 08:50:04.038	Ch1	0x600	标准帧	数据帧	CAN	Rx	8	00 01 00 0B 00 01 00 00
16	2022-04-12 08:50:04.038	Ch1	0x701	标准帧	数据帧	CAN	Rx	8	04 4F 29 F7 80 21 00 02
17	2022-04-12 08:50:04.163	Ch1	0x600	标准帧	数据帧	CAN	Rx	8	00 02 00 0C 00 01 00 00
18	2022-04-12 08:50:04.163	Ch1	0x701	标准帧	数据帧	CAN	Rx	8	04 4F 29 F7 80 21 18 03
19	2022-04-12 08:50:04.163	Ch1	0x701	标准帧	数据帧	CAN	Rx	8	0D 4E D9 FC 80 20 00 08

Figure 2- 1 CAN protocol output

14	2022-04-12 08:54:53.883	Ch1	0x701	标准帧	数据帧	CANFD加速	Rx	64	00 4F 51 F3 7F DF 82 08 01 4F 89 EE 7F DF E8 09 02 50 21 E7 80 1F 48 08 03 4F 59 E7 80 20 80 01
15	2022-04-12 08:54:53.961	Ch1	0x701	标准帧	数据帧	CANFD加速	Rx	64	00 4F 51 F3 7F DF 82 08 01 4F 89 EE 7F DF E8 09 02 50 21 E7 80 1F 48 08 03 4F 59 E7 80 20 80 01
16	2022-04-12 08:54:54.087	Ch1	0x701	标准帧	数据帧	CANFD加速	Rx	64	00 4F 51 F3 7F DF 82 08 01 4F 89 EE 7F DF E8 09 02 50 21 E7 80 1F 48 08 03 4F 59 E7 7F E0 80 01
17	2022-04-12 08:54:54.211	Ch1	0x701	标准帧	数据帧	CANFD加速	Rx	64	00 4F 51 F3 7F DF 82 08 01 4F 89 EE 7F DF E8 09 02 50 21 E7 80 1F 48 08 03 4F 59 E7 7F E0 80 01
18	2022-04-12 08:54:54.335	Ch1	0x701	标准帧	数据帧	CANFD加速	Rx	48	00 4F 91 EE 80 1F A0 09 01 50 21 E7 80 1F 48 08 02 4F 59 E7 E0 88 07 03 4E C2 06 7F E0 00 00
19	2022-04-12 08:54:54.470	Ch1	0x701	标准帧	数据帧	CANFD加速	Rx	48	00 4F 99 ED 80 1F A0 09 01 50 21 E7 80 1F 48 08 02 4F 59 E6 7F E0 28 07 03 4E C2 06 7F E0 00 00
20	2022-04-12 08:54:54.586	Ch1	0x701	标准帧	数据帧	CANFD加速	Rx	48	00 4F 99 ED 7F DE E0 09 01 50 21 E7 80 1F 48 08 02 4F 59 E6 7F E0 68 08 03 4E C2 06 7F E0 00 00

Figure 2- 2 CANFD protocol output

Note: When using the CANFD protocol, when the remainder of the total number of targets divided by 8 is 5 or 7 targets, the protocol will use 0xCC to complete 5 targets to 48 bytes (i.e., the length of 6 targets), and complete 7 targets to 64 bytes (i.e., the length of 8 targets). The example is as follows.

2022-05-27 15:56:18.777	Ch0	0x701	标准帧	数据帧	CANFD加速	Rx	64	00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 84 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 00
2022-05-27 15:56:19.870	Ch0	0x701	标准帧	数据帧	CANFD加速	Rx	48	00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 00
2022-05-27 15:59:06.136	Ch0	0x701	标准帧	数据帧	CANFD加速	Rx	64	00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 00
2022-05-27 15:59:06.633	Ch0	0x701	标准帧	数据帧	CANFD加速	Rx	48	00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 00

Figure 2- 3The number of targets is 13 divided by 8 and the remainder is 5 targets.

2022-05-27 14:23:01.232	Ch0	0x701	标准帧	数据帧	CANFD加速	Rx	64	00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 00
2022-05-27 14:23:01.298	Ch0	0x701	标准帧	数据帧	CANFD加速	Rx	64	00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 00
2022-05-27 14:23:01.395	Ch0	0x701	标准帧	数据帧	CANFD加速	Rx	64	00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 00
2022-05-27 14:23:01.533	Ch0	0x701	标准帧	数据帧	CANFD加速	Rx	64	00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 4E 21 FF 84 E0 00 00 00 00
2022-05-27 14:23:01.597	Ch0	0x701	标准帧	数据帧	CANFD加速	Rx	64	00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 4E 21 FF 89 AD 00 00 00 00

Figure 2- 4There are only seven targets.

2.1 Version information (0x700)

	7	6	5	4	3	2	1	0
0	Soft MajorRelease _{msb}	6	5	4	3	2	1	0
1	Soft MinorRelease _{msb}	14	13	12	11	10	9	8
2	Soft PatchLevel _{msb}	23	22	21	20	19	18	17

Figure 2- 5Version Information (0x700) Memory Distribution Map

Table 2- 2Description of version information (0x700)

Signal	Start	Len	Min	Max	Res	Description
Soft_MajorRelease	0	8	0	255	1	Major version of software
Soft_MinorRelease	8	8	0	255	1	Software minor

						version
Soft_PatchLevel	16	8	0	255	1	Software patch version

Sort_MajorRelease: Major Software Version

Soft_Minor Release: Software Minor Release

Soft_PatchLevel: Software Patch Version

2.2 Point cloud status information (0x600)

This message is sent once per
cycle

	7	6	5	4	3	2	1	0
0	Cluster NofObjects 7 msb	6	5	4	3	2	1	0
1	Cluster NofObjects 15 msb	14	13	12	11	10	9	8
2	Cluster MeasCount 23 msb	22	21	20	19	18	17	16
3	Cluster MeasCount 31 msb	30	29	28	27	26	25	24
4	Interface Version 39 msb	38	37	36	35	34	33	32
5	SubFrame Number 47 msb	46	45	44	43	42	41	40

Figure 2-6Point cloud status information (0x600) memory distribution map

Table 2- 3Description of point cloud status information (0x600)

Signal	Start	Len	Min	Max	Res	Description
Cluster_NofObjects	8	16	0	65535	1	Total number of targets
Cluster_MeasCounter	24	16	0	65535	1	Cycle count
Cluster_InterfaceVersion	32	8	0	255	1	0
Cluster_SubFrameNumber	40	8	0	255	1	Number of subframes

0x600 Parsing example

1) The message is 0 x01 0x20 0 x02 0 x04 0 x00 0 x00 0 x00 0 x00

Namely:

Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
0x01	0x20	0x02	0x04	0x00	0x00	0x00	0x80

2) Number of targets: $0 \text{ x}01 * 256 + 0 \text{ x}20 = 288$;

- 3) Cycle count: $0 \times 02 * 256 + 0 \times 04 = 516$;
- 4) Interface version: 0 by default;
- 5) Number of sub-frames: 0 or 1;

2.3 Point cloud data information (0x701)

This message contains the position and velocity, and outputs the raw point cloud information.



Figure 2-7 Point cloud data information (0x701) memory distribution map

Table 2- 4 Point cloud data information (0x701) description

Signal	Start	Len	Min	Max	Res	Unit
Cluster_ID	0	8	0	255	1	
Cluster_DistLong	19	13	-100	+309.55	0.05	m
Cluster_DistLat	24	11	-50	+52.35	0.05	m
Cluster_VrelLong	46	10	-16.00	15.95	0.05	m/s
Cluster_DynProp	48	3	0	7	1	
Cluster_Class	51	2	0	3	1	Not used yet
Cluster_Height	53	9	-64.00	63.75	0.25	m
Cluster_RCS	56	8	0	255	1	dB

Table 2- 5 Point cloud data information (0x701) description

Signal	Start	Description
Cluster_ID	0	Number of targets
Cluster_DistLong	19	Target longitudinal distance
Cluster_DistLat	24	Target lateral distance
Cluster_VrelLong	46	Target radial velocity

Cluster_DynProp	48	Target type
Cluster_Height	53	Target altitude
Cluster_RCS	56	Target signal to noise ratio

0 x701 Parsing Example

- 1) The message is 0x57 0 x4E 0 xC2 0 x0C 0 x7F 0x60 0x00 0x80

Namely:

Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
0x57	0x4E	0xC2	0x0C	0x7F	0x60	0x00	0x80

- 2) Target ID: 0 x57 = 87, that is, the target ID is 87. The target is generated cyclically between 0 and 255. During stable tracking, the target ID remains unchanged;
- 3) Target longitudinal distance: (0x4E * 32 + 0xC2 >> 3) * 0.05-100 = 26m;
- 4) Target lateral distance: ((0xC2 & 0x07) * 256 + 0x0C) * 0.05-50 = -23.8m;
- 5) Target radial velocity: (0x7F * 4 + 0x60 >> 6) * 0.05-16 = 9.45m/s;
- 6) Target height: ((0x60 & 0 x3F) * 8 + 0 x00 >> 5) * 0.25-64 = 0 m;
- 7) Target dynamic attribute: 0 x00 & 0 x07 = 0, all are 0 by default;
- 8) Target SNR: 0x80 = 128.
- 9) Target radial distance R = sqrt (Cluster _ DistLong * Cluster _ DistLong + Cluster _ DistLat * Cluster _ DistLat)。
- 10) The target angle is θ . Note the difference between the units of angle and radian.

$$\tan \theta = \frac{Cluster_DistLat}{Cluster_DistLong}$$

2.4 Trace Status Information (0x600)

This message is sent once per cycle

	7	6	5	4	3	2	1	0
0	Objects 7 NofObjects msb	6	5	4	3	2	1	0 lsb
1	Objects 15 MeasCount msb	14	13	12	11	10	9	8
2	Objects 23 MeasCount	22	21	20	19	18	17	16 lsb
3	Objects 31 CarSpeed msb	30	29	28	27	26	25	24
4	Objects 39 CarSpeed	38	37	36	35	34	33	32 lsb

Figure 2- 8Point cloud status information (0x600) memory distribution map

Table 2- 6Description of point cloud status information (0x600)

Signal	Start	Len	Min	Max	Res	Description
Objects_NoOfObjects	0	8	0	255	1	Total number of targets
Objects_MeasCounter	16	16	0	65535	1	Cycle count
Objects_CarSpeed	32	16	0	65535	1	Body speed, accuracy 0.1m/s, range [-20m/s, 20m/s]

0x600 Parsing example

1) The message is 0 x21 0 x01 0 x02 0 x00 0 x91 0 x00 0 x00 0 x00

Namely:

Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
0x21	0x01	0x02	0x00	0x91	0x00	0x00	0x80

- 2) Number of targets: 0 x21 = 33;
- 3) Cycle count: 0x01 * 256 + 0x02 = 258;
- 4) Vehicle body speed: (0x00 * 256 + 0x91) * 0.1-20 = -5.5m/s;

2.5 Trace data information (0x701)

This message contains output track status information such as position, velocity, and so on.

	7	6	5	4	3	2	1	0
0	Objects_FrameID 7	Objects ID 6	5	4	3	2	1	0
1	Objects_DistY 15	14	13	12	11	10	9	8
2	Objects_DistY 23	22	21	20	19	Objects_DistX 18	17	16
3	Objects_DistX 31	30	29	28	27	26	25	24
4	Objects_VreY 39	38	37	36	35	34	33	32
5	Objects_VreY 47	46	Objects_VreX 45	44	43	42	41	40
6	Objects_VreX 55	54	53	Objects Class 52	51	Objects DynProp 50	49	48
7	Objects RCS 63	62	61	60	59	58	57	56

Figure 2- 9Trace Data Information (0x701) Memory Distribution

Map

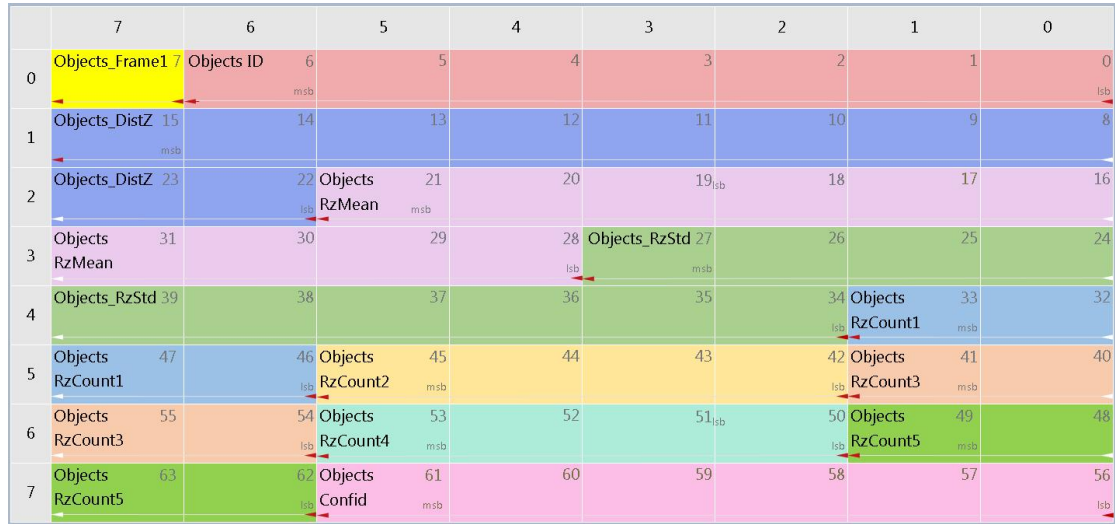


Figure 2-10 Trace data information (0x701) memory distribution map

Table 2-7 Trace data information (0x701) description

Signal	Start	Len	Min	Max	Res	Unit
Objects_ID	0	7	0	127	1	
Objects_Frame0	7	1	0	1	1	The default is 0
Objects_DistY	19	13	-100	+309.55	0.05	m
Objects_DistX	24	11	-50	+52.35	0.05	m
Objects_VrelY	46	10	-128.00	127.75	0.25	m/s
Objects_DynProp	48	3	0	7	1	
Objects_Class	51	2	0	3	1	Not used yet
Objects_VrelX	53	9	-64.00	63.75	0.25	m
Objects_RCS	56	8	0	255	1	dB
Objects_ID	0	7	0	127	1	
Objects_Frame1	7	1	0	1	1	The default is 1
Objects_DistZ	22	10	-30	+72.3	0.1	m
Objects_RzMean	28	10	-30	+72.3	0.1	m
Objects_RzStd	34	10	-30	+72.3	0.1	m
Objects_RzCount1	46	4	0	15	1	
Objects_RzCount2	42	4	0	15	1	

Objects _RzCount3	54	4	0	15	1	
Objects _RzCount4	50	4	0	15	1	
Objects _RzCount5	62	4	0	15	1	
Objects _Confid	56	6	0	64	1	

Table 2- 8Trace data information (0x701) description

Signal	Start	Description
Objects _ID	0	Target ID
Objects _Frame0	7	Target information output frame number, 0 by default
Objects _DistY	19	Target longitudinal distance
Objects _DistX	24	Target lateral distance
Objects _VrelY	46	Target longitudinal velocity
Objects _DynProp	48	Target type
Objects _VrelX	53	Target lateral velocity
Objects _RCS	56	Target echo intensity
Objects _ID	0	Target ID
Objects _Frame1	7	Target information output frame number, 1 by default
Objects _DistZ	22	Target altitude
Objects _RzMean	28	Target average altitude
Objects _RzStd	34	Target altitude variance
Objects _RzCount1	46	Number of times of target in height range (-0.5-0.5) m
Objects _RzCount2	42	Target (+ 0.5-+ 1) m times in height range
Objects _RzCount3	54	Times of target in height range (-1.0--0.5) m
Objects _RzCount4	50	Number of times when the target is in the height range of (+ 1.0-+ 30) m
Objects _RzCount5	62	Number of times when the target is in the height range of (-30--1.0) m
Objects _Confid	56	There is a chance of the target

0 x701 Parsing Example

The first frame message is 0x57 0 x4E 0 xC2 0 x0C 0x7E 0x60 0x00 0x62

Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
0x57	0x4E	0xC2	0x0C	0x7E	0x60	0x00	0x62

- 1) Target ID: 0x57 & 0x7F = 87, that is, target ID is 87. The target is generated cyclically between 0 and 127. During stable tracking, the target ID remains unchanged;
- 2) Information output frame number Frame0: (0x57 & 0x80) > 7 = 0. Two frames 0x701 output a target information, and 0 represents that this frame is the first frame information;
- 3) Target longitudinal distance: (0x4E * 32 + 0xC2 >> 3) * 0.05-100 = 26m;
- 4) Target lateral distance: ((0xC2 & 0x07) * 256 + 0x0C) * 0.05-50 = -23.8m;
- 5) Target longitudinal speed: (0x7E * 4 + 0x60 >> 6) * 0.25-128 = -1.75m/s;
- 6) Target lateral velocity: ((0x60 & 0 x3F) * 8 + 0x00 >> 5) * 0.25-64 = 0m/s;
- 7) Target dynamic attribute: 0 x00 & 0 x07 = 0, all are 0 by default;
- 8) Target echo intensity: 0 X 62 = 98.
- 9) Target radial distance R = sqrt (Objects _ DistY * Objects _ DistY + Objects _ DistX * Objects _ DistX)。
- 10) The target angle is θ . Note the difference between the units of angle and radian.

$$\tan \theta = \frac{Cluster_DistLat}{Cluster_DistLong}$$

The second frame message is 0xD7 0x4E 0xD4 0x15 0x09 0x88 0x04 0x00:

Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15
0xD7	0x4E	0xD4	0x15	0x09	0x88	0x04	0x00

- 11) Target ID: 0 xD7 & 0x7F = 87, that is, the target ID is 87. The target is generated cyclically between 0 and 127. During stable tracking, the target ID remains unchanged;
- 12) Information output frame number Frame1: (0 xD7 & 0x80) > 7 = 1. Two frames 0x701 output one target information, and 1 represents that this frame is the second frame information;
- 13) Target height: (0x4E * 4 + 0 xD4 >> 6) * 0.1-30 = 1.5m;
- 14) Target average height: ((0xD4 & 0x3F) * 16 + 0x15 >> 4) * 0.1-30 = 2.1m;
- 15) Target height variance: ((0x15 & 0 x0F) * 64 + 0x09 >> 2) * 0.1-30 = 2.2m;
- 16) Times of target in height range (-0.5-0.5) m: (0x09 & 0x03) * 4 + (0x88 >> 6) = 6;
- 17) Number of times of target in height range (+ 0.5-+ 1) m: (0x88 & 0x3C) >> 2 = 2;
- 18) Number of times of target in height range (-1.0--0.5) m: (0x88 & 0x03) * 4 + (0x04 >> 6) = 0;

-
- 19) Number of times of target in height range (+ 1.0-+ 30) m: $(0x04 \ \& \ 0x3C) \ \> \ 2 = 1$;
- 20) Number of times of target in height range (-30--1.0) m: $(0x04 \ \& \ 0x03) * 4 + (0 \ x00 \ \> \ 6) = 0$;